INTRODUCTION:

Cancer is a highly complex disease, with significant patient-to-patient variation in both genomic changes and clinical outcomes. Individuals may show noticeably different survival trajectories even within the same cancer type, indicating the existence of underlying latent biological subgroups. Precision medicine advancement and better risk stratification are contingent upon the identification of these subgroups. Clinicogenomic data, which combines molecular and clinical information, has been made possible by recent developments in cancer genomics. It is possible to investigate whether hidden patient structure exists based on combined genomic and clinical features using large-scale cohorts like the MSK-CHORD dataset. These subgroups must be inferred using statistical and machine learning techniques as they are not directly observable.

This study's purpose is to use clinicogenomic data to identify latent patient subgroups and assess whether these subgroups are linked to different survival outcomes. Our specific goals are to: (i) use unsupervised learning techniques to find patient clusters; (ii) describe these clusters using clinical and genomic characteristics; and (iii) evaluate the relationship between cluster membership and overall survival. The goal of this analysis is to determine whether these patterns can guide clinically significant risk stratification and how genomic variation affects patient outcomes.

DATA PROCESSING AND VARIABLES USED:

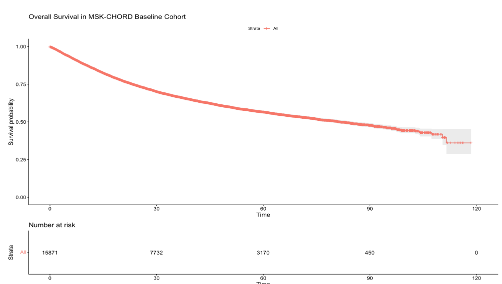
The MSK-CHORD cohort, which contains clinical, genomic and survival data, was used for the analysis. By identifying relevant baseline variables and transforming them for clustering and survival analysis, a patient-level dataset was created.

Clinical, genomic and survival characteristics were among the variables used. Because age is a significant prognostic factor, it was included as a continuous variable. The severity of the disease was measured using the tumor stage, with a focus on advanced disease (Stage 4), which is linked to worse outcomes. TP53, KRAS, APC, EGFR, BRAF, PTEN, and RBM10 are important cancer driver genes for which binary mutation indicators (0/1) were developed. Major biological pathways in cancer are represented by these genes. TP53 is associated with genomic instability; KRAS and BRAF with MAPK-driven proliferation; APC with Wnt signaling; EGFR with receptor-mediated growth; PTEN with PI3K-AKT signaling; and RBM10 with RNA splicing. When combined, they detect a variety of mechanisms that underlie the development of tumors. Overall survival time (OS_TIME) and event indicator (OS_EVENT) were used to define survival outcomes.

Data processing included removing observations with missing critical variables, converting mutation data into binary indicators, and organizing the data with one row per patient. To ensure comparability, continuous variables were standardized before PCA and clustering. About 15,871 patients made up the final dataset, which integrated survival, genomic, and clinical data for further analysis.

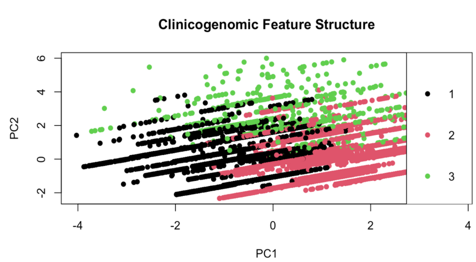
STATISTICAL METHODS :

To identify latent patient subgroups and evaluate their association with survival outcomes, dimensionality reduction, unsupervised clustering, and survival analysis methods were combined and applied. These approaches allow us to uncover underlying structure in high-dimensional clinicogenomic data and assess whether the identified subgroups correspond to meaningful differences in patient outcomes.



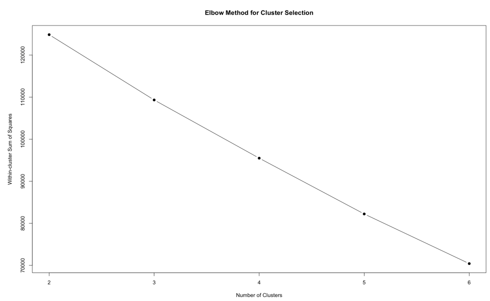
Kaplan–Meier estimation was used to characterize overall survival in the cohort. As shown in Figure 1, survival probabilities decrease steadily over time, with substantial variability observed across patients. This heterogeneity in survival outcomes suggests the presence of underlying latent subgroups, motivating further clustering analysis.

Fig 1. Overall survival (Kaplan–Meier) for baseline cohort



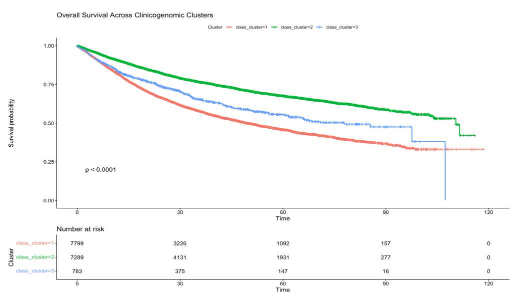
Principal Component Analysis (PCA) was applied to reduce the dimensionality of clinicogenomic features and visualize underlying structure in the data. As shown in Figure 2, patients exhibit structured variation in the reduced feature space, with visible groupings. When colored by cluster assignment, the PCA projection shows partial separation between clusters, indicating that the clustering captures meaningful variation in the data.

Fig 2. Projection of clinicogenomic features



K-means clustering was used to identify latent patient subgroups, with the optimal number of clusters determined using the elbow method. As shown in Figure 3, the within-cluster sum of squares decreases with increasing cluster number, with a noticeable change in slope around k = 3. This suggests that a three-cluster solution adequately captures the major structure in the data.

Fig 3. Elbow Plot for Cluster Selection

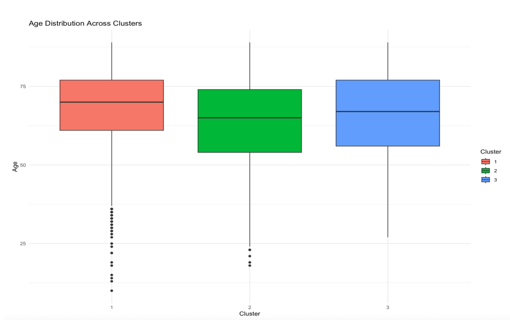


Kaplan–Meier survival curves stratified by cluster membership are shown in Figure 4. Clear separation is observed between the three clusters, indicating substantial differences in survival outcomes. Patients in Cluster 3 (High-Risk) exhibit the poorest survival, while Cluster 1 (Low-Risk) shows the most favorable outcomes, with Cluster 2 (Intermediate-Risk) lying between the two. The difference in survival across clusters is statistically significant ($p < 0.001$), demonstrating that the identified clusters capture meaningful survival risk stratification.

Fig 4. Survival Across Clinicogenomic Clusters

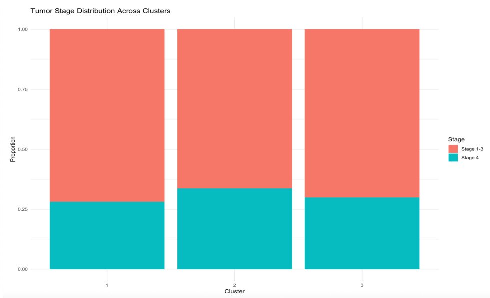
All analyses were conducted using standard statistical and machine learning workflows. Continuous variables were standardized prior to analysis to ensure comparability across features. K-means clustering was performed using Euclidean distance on the standardized feature space, and the selected cluster solution was evaluated based on both statistical criteria and interpretability. Survival differences between clusters were assessed using Kaplan - Meier estimation and compared using the log-rank test, while Cox proportional hazards regression was used to quantify the association between cluster membership and survival risk. Results were interpreted in the context of both statistical significance and clinical relevance

RESULTS :



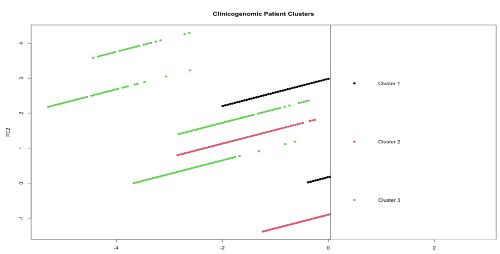
Age distributions across the identified clusters are shown in Figure 5. Moderate differences are observed between clusters, with some clusters exhibiting slightly higher median age than others. However, the differences are not extreme, suggesting that age alone does not fully explain the clustering structure. This indicates that genomic features play a key role in defining the identified subgroups.

Fig 5. Age Distribution Across Clusters



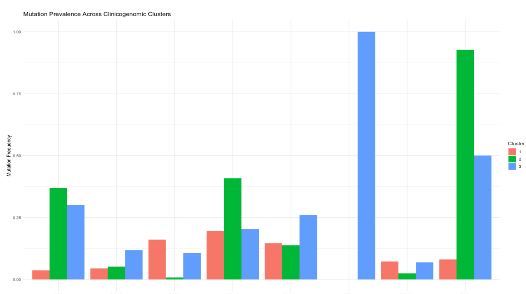
The distribution of tumor stage across clusters is shown in Figure 6. Differences in stage composition are observed, with certain clusters containing a higher proportion of advanced-stage (Stage 4) disease. In particular, clusters associated with poorer survival outcomes tend to have a greater representation of advanced-stage tumors. These findings suggest that the identified clusters capture clinically relevant differences in disease severity.

Fig 6. Tumor Stage Distribution Across Clusters



The PCA projection of clinicogenomic features is shown in Figure 7, with points colored by cluster assignment. The plot reveals structured variation in the data, with partial separation between clusters in the reduced feature space. While overlap exists, the observed grouping suggests the presence of latent subgroups captured by the clustering algorithm.

Fig 7. PCA Visualization of Cluster Assignments



Mutation prevalence varies substantially across the identified clusters, as shown in Figure 8. Cluster 2 exhibits a high frequency of TP53 mutations, suggesting increased genomic instability, while Cluster 3 is characterized by a high prevalence of PTEN mutations, forming a distinct genomic subgroup. Differences are also observed in other driver genes such as KRAS and APC. These distinct mutation patterns support the interpretation that the clusters represent biologically meaningful subgroups rather than arbitrary

Fig 8. Mutation Prevalence Across Clusters

Overall survival analysis revealed a substantial heterogeneity in patient outcomes within the patients identified. Unsupervised clustering of clinicogenomic features identified three distinct patient groups, which showed partial separation in the PCA projection, indicating underlying structure in the data. Kaplan–Meier analysis demonstrated clear and statistically significant differences in survival across clusters, with Cluster 3 (High-Risk) exhibiting the poorest outcomes, Cluster 1 (Low-Risk) showing the most favorable survival, and Cluster 2 (Intermediate-Risk) lying between the two. Further examination of genomic profiles revealed distinct mutation patterns across clusters, including high TP53 prevalence in one subgroup and enrichment of PTEN mutations in another. Differences in clinical characteristics, such as age distribution and tumor stage composition, were also observed across clusters. Together, these findings indicate that the identified clusters represent meaningful clinicogenomic subgroups that capture both biological variation and clinically relevant survival differences.

DISCUSSION :

This study's findings indicate that the combination of clinical and genomic characteristics provides insights into significant underlying structures among cancer patient groups. The clusters identified display notable variations in survival rates, as evidenced by Kaplan–Meier analysis and genomic mutation data. These results are suggestive that clinicogenomic clustering can identify biologically unique subgroups that hold clinical significance, suggesting its potential to enhance risk stratification in cancer care. We observe that the clustering revealed an increased presence of essential driver mutations in particular groups, which points to the possibility that fundamental molecular processes may possibly influence variations in patient outcomes.

While these results provide valuable insights, several limitations must be acknowledged. First, the study focused on a restricted selection of genes rather than encompassing the entire mutational landscape, potentially hindering the detection of more intricate genomic patterns. Second, the clustering outcomes rely on the choice of features and the application of k-means clustering, which presupposes a spherical structure of clusters and may not accurately reflect the genuine biological diversity. Third, the survival associations identified in this research are observational and do not establish causal links. Furthermore, the lack of external validation with independent datasets restricts the broader applicability of the results.

Future research will aim to broaden this analysis by integrating a wider array of genomic features, including pathway-specific mutation classifications, to enhance our understanding of the underlying biological processes. We will investigate cluster patterns within individual cancer types to determine if analogous subgroup structures and survival associations emerge across various tumor contexts. Subsequently, we plan to explore alternative clustering techniques, such as model-based approaches, to refine and compare subgroup identification. Additional analyses will involve gene-level survival modeling to assess the influence of significant mutations on patient outcomes. Lastly, these results will be tested against independent cancer genomics cohorts to examine their reproducibility and evaluate their potential use in clinical risk stratification.